

445. (continued)

A MEMOIR ON QUARTIC SURFACES.

(continuation, articles 60–112)

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Dianodal Curve of 8 Points.

60. Consider first any 8 points whatever 1, 2, 3, 4, 5, 6, 7, 8; the dianodal surface 1234567, our 7 on the dianodal surface 1234568. The two surfaces have in common the line through the points 1, 2, 3, 4, 5, 6, twice; this would imply a reduction 25, viz. we should have $5 \cdot 6 = 30 - 25$. It is not understood this, and I drop the question.

61. However, the line 12 is in involution 6 given quadrics, there should be a reduction 23, viz. we should have

$$(a, b, c, d, f, g, h, l, m, n \times P, Q, R, S, T, U, V, W, X, Y)^2 = 0$$

of a fourth quadric surface to touch the line, in order that the dianodal curve $5 \cdot 6 = 30 - 23$.

62. It may be observed, in regard to the foregoing solution, that if we take

$$123 = 0, \quad 456 = 0, \quad 78 = 0,$$

the line 12, ... in involution with the 6 given quadrics; the relation between $(a, b, c, d, \dots)$, 4 equations; and similarly, if we attempt to find m_1, m_2, m_3, m_4 , such that

$$m_1(P_1 Q_1, \dots) + m_2(P_2 Q_2, \dots) + m_3(P_3 Q_3, \dots) + m_4(P_4 Q_4, \dots) = 0;$$

we have between the coefficients (b, c) 4 equations; and similarly, if we attempt to find m_1, m_2, m_3, m_4 such that

$$\omega_1(1, b_1, c_1 \times P_1, Q_1)P + \dots + \omega_4(1, b_4, c_4 \times P_4, Q_4)P = 0,$$

we have between the coefficients (b, c) 4 equations; viz., these may be determined and the system give 6 lines for another given axis; we have more equations between the coefficients (b, c) ; viz., these will be determined by 5 equations of the form (b, c) 4 equations; and the result should be of this nature, that if at any point of the axis a quadric were to pass through 4 of the given lines, the quadric surface generated by the lines we have a close cubic meeting the axis 3 times.

63. It may be observed, in working this out, to consider H, K as standing, in the first instance, for $(x, y, z, w), (x', y', z', w')$, three random points respectively. Writing also, for shortness, take the abovementioned points H, H' to denote the points on the absolute conic of one of the 4 quadrics cutting harmonically the system of the 4 points, Q on the symmetroid, P and P' on the Jacobian, Q' on the symmetroid; but in what follows I make no use of this last point Q' .

The Symmetroid: Symmetroid and Jacobian.

I have established in the foregoing general theory; but it is a particular case of it which connects itself with the theory of nodal quartics; viz., the cube of coefficients is a symmetrically arranged cube

$$\begin{array}{cccc} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & d \end{array}$$

or say its upper face is the symmetrical square matrix

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix}$$

and the other horizontal planes, the like square with the several letters affected by suffixes.

The surface $\nabla = 0$ is here a surface of the form

$$\nabla = \begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

($A, B, \&c.$ linear functions of (x, y, z, w)), viz., ∇ is a symmetrical determinant; I call this a symmetroid; the surface $\nabla = 0, \Theta = 0$ are not the same, and the same point, and the correspondence R to P is a reciprocal one; or that, instead of the Jacobian being the locus of points, we have only 1 point on the surface ∇ , and 2 points P, P' on the surface $\Theta = 0$; viz., the diagram is

$$\begin{array}{c} \Delta, \Theta, 0, 0; 0, 0, \dots, \Delta \\ \dots, Q, P, Q, P, P', Q, \dots \end{array}$$

moreover the symmetroid surface $\nabla = 0$ is a surface with 10 nodes, which is clearly not octadic, and which is therefore the decadianome.

The Decadianome.

64. The decadianome has thus 10 nodes; and it is a surface of the form

$$(a, b, c, d, f, g, h, l, m, n \times P, Q, R, S)^2 = 0,$$

where $\Delta = 0$ is a particular quartic surface having the 8 nodes, it at once appears that the 9th node must be on the dianodal surface of the 8 points $\nabla J(P, Q, R, S) = 0$, and on the Jacobian; and then the formulae No. 5, this is a curve of the order 16; the node may be any point whatever on this curve, and taking it to be a given point on the curve, the number of constants in the resulting equation should be 2. Hence if $P = 0$ be the quadric surface through the 9 points, and $\Delta = 0$ is a particular quartic surface having the 9 nodes as nodes, the general equation is $aP^2 + \Delta\Phi = 0$.

65. But we may consider the question somewhat differently. Starting from the equation

$$(a, b, c, d, f, g, h \times P, Q, R, S, T, U)^2 = 0,$$

also on the dianodal surface 1234567; it is clear that 9 must be on the dianodal surface 1234568; the complete intersection is of the order 36, and we have to consider how this breaks up so as to contain as part of itself the dianodal curve of the order 18.

Dianodal Curve of 9 Points.

66. Consider first any 9 points whatever 1, 2, 3, 4, 5, 6, 7, 8, 9; the dianodal surface 1234567, our 7 on the dianodal surface 1234568. The two surfaces have in common the line through the points 1, 2, 3, 4, 5, 6 twice; this would imply a reduction $4 \cdot 10 = 40$, but it is clear that the line 123 is in involution to the dianodal surface, the complete intersection is of the order 36, and we have therefore by what precedes the value $4 \cdot 10 - 18 = 22$, viz. the curve we are in search of is of the order 22.

Pseudosymmetrical Cyzo: the Hessian of a Cubic.

67. Reverting to the general equation

$$aS + \beta T + \gamma U + \delta V = \text{cone},$$

which connects the symmetroid and Jacobian, it is evident that if S, T, U, V are the derivatives, in regard to the coordinates, of a single cubic function $U_c = (* \times x, y, z, w)^3$, then the symmetroid and the Jacobian become one and the same surface; viz., this is the Hessian surface $H = 0$ derived from the given cubic surface. The two corresponding points P on the surface have a coincidence in this case; and the equation $\Phi = 0$ vanishes identically when S, T, U, V are derivatives of a single cubic function. Hence, in either case, we may use as a node on the symmetroid, viz., the Hessian of the cubic surface, of which the equation is

$$xyzw \left(\frac{\alpha}{x} + \frac{\beta}{y} + \frac{\gamma}{z} + \frac{\delta}{w} \right) = 0,$$

a form which puts in evidence the properties just referred to.

Quartics with 11 or more Nodes.

68. I mention two results which, although they relate to quartic surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \dots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has a node $H = 0, G = 0, L = 0$; or the surface has in all 11 nodes. And so also if (identically in every case) B is $= 0$, there are 12 nodes; if C is $= 0$, there are 13 nodes; and if D is $= 0$, 14 nodes respectively, mentioned in Kummer's Memoir with 11, 12, 13, and 14 nodes respectively.

69. We may consider the symmetroid derived from the quartic surfaces which pass through 6 given points; viz., taking as before (see No. 25) the coordinates of the 6 points, the dianodal surface of these 6 is 7, on the dianodal surface 1234568, the surface has 11 nodes; viz., we have, in fact, quartic surfaces with 11, 12, 13, and 14 nodes respectively. But I reserve this for future consideration.

Two Nodes.

70. Suppose there is a 10th node, say 10; starting from the equation $aP^2 + \Delta\Phi = 0$ ($P = 0$ the quadric surface through the 9 points, $\Delta = 0$ a particular quartic surface having the 9 points), we have to express that the surface has 10 also for a node; the conditions are, that on the curve $P = 0, \Delta = 0$, we have

$$aP^2 + \Delta\Phi = 0, \quad a(P^2)_x + (\Delta\Phi)_x = 0, \quad \&c.,$$

or, as this is more conveniently written,

$$(\alpha \dots) \chi(x, y, z, w) = 0;$$

if for a moment $(\alpha, \dots)$ denote the coefficients of the quadric function $aP + \beta T + \gamma U + \delta V$, then $(P, P', \dots)$ are at the point 10.

The Lineodianic Correspondence of Quartic Surfaces.

71. Consider the quartic surfaces

$$S = 0, \quad T = 0, \quad U = 0, \quad V = 0,$$

lineliner in regard to the two sets of coordinates (x, y, z, w) and $(\alpha, \beta, \gamma, \delta)$; viz., each of these expressions is of the form

$$(* \times y, z, w \times \beta, \gamma, \delta).$$

This implies that the point (x, y, z, w) lies on a certain quartic surface $\Theta = 0$, and the point $(\alpha, \beta, \gamma, \delta)$ on a certain quartic surface $\Theta = 0$, and that the two surfaces correspond point to point to each other. In fact, writing the four equations in the form

$$\begin{aligned} La + Mb + Nc + Pd &= 0, \\ L'a + M'b + N'c + P'd &= 0, \\ L''a + M''b + N''c + P''d &= 0, \\ L'''a + M'''b + N'''c + P'''d &= 0, \end{aligned}$$

where $L, \&c.$ are linear functions of (x, y, z, w) , we obtain the equation

$$\Theta = \begin{vmatrix} L, & M, & N, & P \\ L', & M', & N', & P' \\ L'', & M'', & N'', & P'' \\ L''', & M''', & N''', & P''' \end{vmatrix} = 0;$$

and similarly, writing the four equations in $(\alpha, \beta, \gamma, \delta)$ we obtain

$$\begin{vmatrix} A\alpha + B\beta + C\gamma + D\delta = 0, \\ A'\alpha + B'\beta + C'\gamma + D'\delta = 0, \\ A''\alpha + B''\beta + C''\gamma + D''\delta = 0, \\ A'''\alpha + B'''\beta + C'''\gamma + D'''\delta = 0, \end{vmatrix}$$

where $A, \&c.$ are linear functions of $(\alpha, \beta, \gamma, \delta)$, we obtain the equation

$$\Theta = \begin{vmatrix} A, & B, & C, & D \\ A', & B', & C', & D' \\ A'', & B'', & C'', & D'' \\ A''', & B''', & C''', & D''' \end{vmatrix} = 0.$$

and similarly, Δ being $= 0$, the four linear equations in (x, y, z, w) are equivalent to three equations, and give for instance (x, y, z, w) proportional to determinants formed with the matrix

$$\begin{vmatrix} A, & B, & C, & D \\ A', & B', & C', & D' \\ A'', & B'', & C'', & D'' \\ A''', & B''', & C''', & D''' \end{vmatrix}$$

which establishes the point-to-point correspondence of the two surfaces.

72. It would at first sight appear that any quartic surface $(a, b, c, d, f, g, h \times P, Q, R, S, T, U, V)^2 = 0$ might have its equation expressed in the foregoing determinant form $\Delta = 0$. This equation seems, in fact, to contain homogeneously as many as 64 constants. But if we multiply the determinant first into line by a constant determinant, the equation will really contain $64 - (2 \cdot 16 - 1) = 32$ constants; and in order that the quartic surface $(a \times P, Q, R, S, T, U, V)^2 = 0$ may have its equation expressible in the form $\Delta = 0$, a single relation must hold good among the coefficients: but this is passing(?).

73. Returning to the quartic surface

$$\Delta = \begin{vmatrix} A, & B, & C, & D \\ A', & B', & C', & D' \\ A'', & B'', & C'', & D'' \\ A''', & B''', & C''', & D''' \end{vmatrix} = 0,$$

we may convert this not only with the foregoing surface $\Theta = 0$, but in a similar manner with another quartic surface $\Theta = 0$, viz., taking the new coordinates $(\xi, \eta, \zeta, \omega)$, we may form the lineliner equations

$$\begin{aligned} A\xi + B\eta + C\zeta + D\omega &= 0, \\ A'\xi + B'\eta + C'\zeta + D'\omega &= 0, \\ A''\xi + B''\eta + C''\zeta + D''\omega &= 0, \\ A'''\xi + B'''\eta + C'''\zeta + D'''\omega &= 0, \end{aligned}$$

which, by the elimination of $(\xi, \eta, \zeta, \omega)$, gives $\Delta = 0$; and by the elimination of (A, B, C, D) a different determinant, $\Theta = 0$, which is to the points (x, y, z, w) for a cone, or say for shortness,

$$aS + \beta T + \gamma U + \delta V = \text{plane-pair};$$

viz., the values of $(a, \beta, \gamma, \delta)$, which satisfy this condition, belong to a threefold relation between the coefficients of aS , having previously explained location, we may say—intervene to a given point of the symmetroid, and that the line on the Jacobian is the axis of the plane-pair.

74. I have a fish lay 4 cuspe, with then to the plane-pair implies a threefold relation between aS , having previously explained location, but the required number of nodes is equal to the order of the threefold relation; hence the order of $aS + \beta T + \gamma U + \delta V$ is equal to the order of the threefold relation; and the number of solutions would be equal to the order of the threefold relation; and the number of solutions would be equal to the order of $(* \times \dots)^4$, viz., as a further abbreviation referring aS , instead of $(\alpha) aS$, to denote the abovementioned determinants, the equation is

$$\begin{aligned} & ab\,abc + a\,abc - ach + bch - bda - dab\,c \\ & + b'\,abc \cdot bgh - h\,bcg + b\,bcf - ach + bch\,c - bda - dab\,c \\ & + b'\,abc \cdot bgh - h\,bcg + b\,bcf - ach + bch\,c - bda - dab\,c \\ & + \dots \end{aligned}$$

The final result contains 184 terms; viz., as a further abbreviation referring abc , instead of (abc) abc , to denote the abovementioned determinants, the equation is symbolically given by

$$abc \cdot abg - hfg \cdot abc + bcg \cdot abf - acf \cdot bch + chg \cdot abd = 0.$$

75. The final result contains 184 terms; viz., 6 terms in the multiplier abc of aS , T , U , V , W , viz., as a further abbreviation referring abc , instead of (abc) abc , to denote the abovementioned determinants, the equation is symbolically given by

$$acf \cdot bgh - dh\,bcg + a\,cdf - bf\,ach = 0.$$

I am not sure whether the surface of the eighth order has been anywhere considered.

Symmetroid with given Nodes.

76. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form such a symmetroid, we obtain, in answer to put it differently: any 7 points must be the points of a single system; if 8 nodes are required, then it is necessary that the exterior region contain $\Delta\Delta$ a posteriori identification with the decadianome, it would be necessary to construct a $\Delta\Delta$ relation containing 7 given points, and a 9th point anywhere on the dianodal curve; but I have not succeeded in effecting this.

Symmetroid with given Lines.

77. To find the equation to the quadric surface through the three lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), \quad (a_2, b_2, c_2, f_2, g_2, h_2), \quad (a_3, b_3, c_3, f_3, g_3, h_3).$$

Take for the equation to the quadric surface through the three lines

$$\begin{vmatrix} x^2, & y^2, & z^2, & w^2, & xz, & xy, & xw, & yw, & zw \\ a_1^2, & b_1^2, & c_1^2, & f_1^2, & g_1^2, & h_1^2, & a_1g_1, & b_1f_1, & c_1h_1 \\ 2a_1^2, & 2b_1^2, & 2c_1^2, & 2f_1^2, & 2g_1^2, & 2h_1^2, & a_1g_1 + a_1c_1, & b_1f_1 + b_1c_1, & c_1h_1 + c_1d_1 \\ \vdots & & & & & & & & \end{vmatrix} = 0;$$

and if we form thus a determinant with three of its lines relating to the line 1, three of them to the line 2, and three to the line 3, we have the equation of the quadric surface through the three lines $(a, b, c, f, g, h)_1, (a, b, c, f, g, h)_2, (a, b, c, f, g, h)_3$. Take to denote the equation; the order of the quadric surface through the three lines becomes indeterminate (in fact, the plane of the intersecting lines, and throwing out these factors, the condition should be of the order 2 in regard to the coordinates of each line; and throwing out these factors, the condition becomes of the order 2; viz., we have the condition for this is

$$\begin{vmatrix} \dots, & 12, & 13, & 14, & 15 \\ 21, & \dots, & 23, & 24, & 25 \\ 31, & 32, & \dots, & 34, & 35 \\ 41, & 42, & 43, & \dots, & 45 \\ 51, & 52, & 53, & 54, & \dots \end{vmatrix} = 0,$$

being, as it should be, of the order 2 in regard to the coordinates of each line.

Condition that 7 given lines shall lie on a Quartic Surface.

78. Taking the lines to be $(a_1, b_1, c_1, f_1, g_1, h_1), \dots, (a_7, b_7, c_7, f_7, g_7, h_7)$, then in precisely the same way we form a determinant of the order $\frac{1}{2}5 \cdot 4 = 10$ (in regard to the columns of each line, and 4 (in regard to the columns of each line), with the several factors $(\dots, \cdot, \cdot, \cdot, \dots)$ in regard to the coordinates of each line.

The Jacobian of 6 points.

79. Any 6 points whatever may be regarded as points on a skew cubic; and the coordinates (x, y, z, w) may be such, $b_1, \dots, b_5$ are linear functions of (x, y, z, w) , and the coordinates (x, y, z, w) are

$$\begin{vmatrix} x & y & z \\ y & z & w \\ b_1 & b_2 & b_3 \end{vmatrix} = 0.$$

This being so, the coordinates of the 6 points through the skew cubic are taken to be $(1, b_1, c_1, c'_1), (1, b_2, c_2, c'_2), \dots, (1, b_5, c_5, c'_5)$; and the equation of the Jacobian surface of the 6 points is the

80. We may now consider the symmetroid derived from the quartic surfaces which pass through 6 given points; viz., taking as before (see No. 25) the coordinates of the 6 points, putting in evidence the passage of the surface through the skew cubic, viz., writing

$$p_1 = \Sigma\tau_1, \quad p_2 = \Sigma\tau_2, \quad p_3 = \Sigma\tau_3, \quad p_4 = \Sigma\tau_4, \quad p_5 = \Sigma\tau_5;$$

moreover,

$$\Sigma\Sigma' = \frac{1}{6}(7xyzw - 4xx' - 4yy' + 2pp' - aba'b'),$$

and therefore

$$\begin{aligned} L_1 &= \dots - xx' - \Sigma x + 2yzw, \\ L_2 &= \dots - 4xx' - 6yyz + 6yzw, \\ L_3 &= \dots - 4xx' - 6yyz + 6yzw, \\ L_4 &= \dots - xx' - 2y' + 4yzw; \end{aligned}$$

then the equation of the Jacobian surface is

$$\begin{vmatrix} 3(p_1 & +xy, & -2y, & \frac{1}{2}\square) \\ +(& 2pp, & -xx, & \frac{1}{2}\square) \\ +(-xy, & -3p, & , & \frac{1}{2}\square) \\ +3(2xy & -xx, & -xyz, & 6) \end{vmatrix} = 0.$$

There is not much difficulty in the direct investigation; but a simple verification may be obtained by showing that the surface contains upon it the 15 lines 12, 13, ..., 56. Write in the equation

$$(x, y, z, w) = (1, b, c, c'),$$

the value L_1 becomes = ..., the value L_2 becomes = ..., the value L_3 becomes = ..., the value L_4 becomes = ...; and the equation becomes

$$\frac{1}{4}(\sigma\pi - \sigma'\pi')(a - 3a' + 3(2(p_2 - p_3))) \cdot \frac{1}{2}(c - c'\pi + \frac{1}{2}c\pi') = 0;$$

viz., relations of the form $\sigma + cb + c = 0$ at the vertices, and the which is the equation are

$$\begin{aligned} -2\Sigma(1, p_1, p_2, p_3, p_4 \times \sigma_1, \pi_1) &= 0, \\ 2\Sigma(1, p_1, p_2, p_3, p_4 \times \sigma_1, \pi_1) \cdot 7 &= 0, \end{aligned}$$

viz., this equation is satisfied if σ denote any one of the quantities $(1, b_1, c_1, c'_1)$, and 1 any one of the same 4 quantities; that is, the equation of the surface is satisfied when (x, y, z, w) are the coordinates of a point on the line joining any 2 of the 6 points.

Locus of the vertex of a Quadric Cone which touches each of Six given Lines.

81. Representing as before each line by means of its six coordinates, let (x, y, z, w) be the coordinates of the vertex, and (X, Y, Z, W) current coordinates. Suppose this (a, b, c, f, g, h) are the coordinates of any one of the lines, the equation of the plane through this line and the vertex is

$$\begin{aligned} & a(wY - wZ) + b(yW - wY) + c(zX - xZ) \\ & + f(yZ - zY) + g(zX - xZ) + h(xY - yX) = 0; \end{aligned}$$

or, what is the same thing, writing for shortness

$$P = \dots, \quad by - gz + dw,$$

$$Q = -hx, \quad +fz - bw,$$

$$R = gx - fy, \quad +cw,$$

$$S = -ax - by - cz,$$

the equation is

$$PX + QY + RZ + SW = 0;$$

and forming in like manner the equations belonging to each of the given lines, the condition that the 6 cones may unite into a single cone, is

$$(\sigma, b, c, f, g, h) \begin{vmatrix} P & Q & R \end{vmatrix}^2 = 0,$$

where the left-hand side represents a determinant of 6 lines, the several lines being respectively $(P, Q, R)_1, (P, Q, R)_2, \dots, (P, Q, R)_6$, also S denoting that the equation by

$$(P, Q, R)^2 = 0.$$

To ascertain the form of this, write for a moment $y = 0, z = 0$; the equation becomes $\{(au, -hx + dw, gx - fx)^2 = 0\}$; or attending only to the highest and lowest powers of x , this is

$$x^2(a, a, -h, g)^2 = 0,$$

and it is thence easy to infer that the whole equation divides by w^4 , we thus, omitting this factor, the form of the equation is

$$(\sigma, b, c, f, g, h) \begin{vmatrix} x & y & z & w \end{vmatrix}^4 = 0,$$

viz., the equation is satisfied if x denote any one of the 6 coordinates (a, b, c, f, g, h) , and 1 any one of the same 6 quantities (a, b, c, f, g, h) ; that is, the equation is satisfied when (x, y, z, w) are the coordinates of any one of the 6 lines, and the degree in the coordinates (x, y, z, w) is 4. It would not be very easy to find the value of the discriminant of the equation.

Quartic Surface through three given Lines.

82. To find the equation of the quartic surface through the three lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), \quad (a_2, b_2, c_2, f_2, g_2, h_2), \quad (\alpha', \beta', \gamma', \delta'),$$

we may form the determinant

$$\begin{vmatrix} x^4 & y^4 & z^4 & w^4 & yz & xz & xy & xw & yw & zw \\ x^3y & xy^3 & yz^3 & y^3z & \dots & \dots & \dots & \dots & \dots & \dots \\ 2x^3 & 2y^3 & 2z^3 & 2w^3 & a\beta' & a\gamma' & a\delta' + a\beta'\gamma' & b\gamma' + a\beta'\delta' & \dots & \dots \\ a^4 & \beta^4 & \gamma^4 & \delta^4 & \beta\gamma & a\beta & a\delta & a\gamma & \beta\delta & \gamma\delta \\ a'^4 & \beta'^4 & \gamma'^4 & \delta'^4 & \beta'\gamma' & a'\beta' & a'\delta' & a'\gamma' & \beta'\delta' & \gamma'\delta' \end{vmatrix} = 0;$$

and if we form thus a determinant with three of its lines relating to the line 1, three to the line 2, and three to the line 3, we have the equation of the quartic surface through the three lines.

Condition that five given lines may lie on a Cubic Surface.

83. Taking the lines to be $(a_1, b_1, c_1, f_1, g_1, h_1), \dots, (a_5, b_5, c_5, f_5, g_5, h_5)$, and (X, Y, Z, W) the coordinates of the vertex, the equation of the cubic surface through the 5 lines is taken to be $(1, b_1, c_1, c'_1), \dots, (1, b_5, c_5, c'_5)$; and the equation of the Jacobian surface of the 5 points

and hence it appears that, forming a determinant of 20 lines, wherein four lines relate to the four required condition. But the condition so obtained is of the order $(\frac{1}{2} \cdot 4) = 6$ in regard to the coordinates of each line; and, as for the quadric, it is satisfactorily contains the several factors which lead to zero, we leave the required relation between the five lines that they shall all of them be cut by 4 or 5 lines; and moreover that, writing $a_i f_i + b_i g_i + c_i h_i = 0$, &c., then every one of the diagonal determinants of the order 4 vanishes; the condition for this is

$$\begin{vmatrix} \cdot, & 12, & 13, & 14, & 15 \\ 21, & \cdot, & 23, & 24, & 25 \\ 31, & 32, & \cdot, & 34, & 35 \\ 41, & 42, & 43, & \cdot, & 45 \\ 51, & 52, & 53, & 54, & \cdot \end{vmatrix} = 0,$$

being, as it should be, of the order 2 in regard to the coordinates of each line.

Condition that 7 given lines shall lie on a Quartic Surface.

84. Taking the lines to be $(a_1, b_1, c_1, f_1, g_1, h_1), \dots, (a_7, b_7, c_7, f_7, g_7, h_7)$, then in precisely the same way we form a determinant of the order 7, which sufficiently shows the conditions; the order of the surface is 4, and so similarly any number of lines on a given Quartic Surface have many to be reduced 4; the more the equation it would be in fact be needed.

General result as to the Dianomes.

85. We have thus established the theory of the dianome quartic surfaces; viz., we have

- The octadianome, 8 nodes, 7 of them arbitrary, and the 8th on an arbitrary line passing through it.
- The enneadianome, 9 nodes, the 9th on the dianodal curve of the 8 points.
- The decadianome, 10 nodes, the 10th any one of the 22 [13] dianodal points of the 9 points.

And as already mentioned, so long as the first 7 nodes are arbitrary, there cannot be more than 10 nodes in all.

The Jacobian with given Lines.

86. The Jacobian contains 24 constants; obviously it is uniquely determined if 4 of the plane-pairs thereof are given; and it is also uniquely determined if 4 of the nodes of the symmetroid can be considered as corresponding to given plane-pairs, or lines of the Jacobian. Take as given any 4 nodes of the symmetroid: viz., if 4 of the plane-pairs are given, the corresponding 4 plane-pairs are taken to be given plane-pairs; and, for the corresponding 4 nodes of the symmetroid would be considered as given, viz., 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1; the given plane-pairs $P_1Q_1 = 0$, $P_2Q_2 = 0$, $P_3Q_3 = 0$, $P_4Q_4 = 0$ of values of values such that we have

$$(S, T, U, V) = (1, 0, 0, 0)P_1Q_1 + (0, 1, 0, 0)P_2Q_2 + (0, 0, 1, 0)P_3Q_3 + (0, 0, 0, 1)P_4Q_4 = \text{plane-pair};$$

and $(1, 1, 1, 1)$ is of the symmetroid; we have to assume

$$(S, T, U, V) = (l_1 P_1Q_1 + l_2 P_2Q_2 + l_3 P_3Q_3 + l_4 P_4Q_4 = \text{plane-pair}),$$

and $(1, 1, 1, 1)$ is of the symmetroid, we have to assume

$$(S, T, U, V) = (l_1 P_1Q_1, l_2 P_2Q_2, l_3 P_3Q_3, l_4 P_4Q_4) = \text{plane-pair of given axis.}$$

87. Suppose, however, that on the Jacobian we have given, not 4 plane-pairs, but 6 lines lying $(l_1, l_2, l_3, l_4, l_5, l_6)$. As before $P_1Q_1 = 0, \dots, P_4Q_4 = 0$ are the first instance undetermined. If we attempt to find $l_1, l_2, l_3, l_4, l_5, l_6$, any one of these lines, l_4 , can be a plane-pair of values such that we have

$$l_1 P_1Q_1 + l_2 P_2Q_2 + l_3 P_3Q_3 + l_4 P_4Q_4 = \text{plane-pair of given axis,}$$

we have between the coefficients (l, l_1, l_2, l_3, l_4) 4 equations; and similarly, if we attempt to find m_1, m_2, m_3, m_4 , such that

$$m_1 P_1Q_1 + m_2 P_2Q_2 + m_3 P_3Q_3 + m_4 P_4Q_4 = \text{plane-pair of another given axis,}$$

we have more equations between the coefficients (b, c) ; viz., these will be determined by 6 equations of the form (b, c) 4 equations; and the result should be of this nature, that on the axis of the assumed plane-pair, the cubic should be in 6 nodes on the surface generated by the lines we have $S = c\xi T$ — at axis 3 times.

Correspondence in the Jacobian: Lines and Skew Cubics.

88. I consider the correspondence of two points on the Jacobian; it is to be shown that when one of the points is on a line of the Jacobian, the corresponding point will be on it. Call the plane-pairs of the system of quadric surfaces 1, 2, 3, ..., 10; selecting any 4 of these (on the dianodal surface of the system of quadric surfaces 1, 2, 3, ..., 10), the corresponding 6 nodes (on the symmetroid) will be on a plane on this plane, the equation of the plane through these would be required correspondence. Conversely, if we have an axis of one of the plane-pairs, say of the plane pair 1, the point P , in the plane pair 1 in regard to these 4 plane-pairs will meet in a point; and observe that, in regard to any 4 plane-pairs, the polar plane of a point in regard to these 4 plane-pairs meet in a point. Call this point P' ; moves along the line, the corresponding point moves along a curve, viz., in particular, the corresponding curve to a skew cubic; that is, the corresponding curve to a line of the Jacobian is a skew cubic, cutting plane-pair twice. In particular, the corresponding curve to a line of the Jacobian; we will have therefore 4, 5, ..., 10 each have a corresponding skew cubic, cutting the line 1, 2; on it; we have therefore 6 skew cubics cutting the line 1, 2 of the Jacobian.

89. I cut a moment to prove the subsidiary theorem assumed in the foregoing investigation. Let the 3 plane-pairs be $PQ = 0$, $RS = 0$, $TU = 0$, the line joining the points (x, y, z, w) and (x', y', z', w') ; the coordinates of any point on the line are $(\lambda x + \mu x_1, \lambda y + \mu y_1, \lambda z + \mu z_1, \lambda w + \mu w_1)$; and hence the equation may be written as

$$[(\lambda x + \mu x_1) \&c. \dots] PQ + (\lambda x + \mu x_1) \&c. RS = 0;$$

viz., the equation may be written as

$$\lambda(P, Q + \mu(P, Q + \dots)) = 0;$$

forming the like equations in regard to $RS = 0$ and $TU = 0$, and eliminating λ, μ , we obtain the required locus

$$\begin{vmatrix} P_1 Q_1 + PQ, & R_1 S_1 + RS, & T_1 U_1 + TU \\ P_1 Q + PQ_1, & R_1 S + RS_1, & T_1 U + TU_1 \end{vmatrix} = 0,$$

forming the like equations in regard to RS and TU , and eliminating λ, μ , we obtain the required locus

$$\begin{vmatrix} (P Q_1 + Q P_1) \delta_{1\nu} \cdot \delta_{1\mu} + (P, Q_1 \delta_{1\nu} \cdot \delta_{1\mu}), & (\dots), & (\dots) \\ \cdot \delta_{1\nu} \cdot R P, & \cdot \delta_{1\nu} \cdot S P, & \cdot \delta_{1\nu} \cdot T P \end{vmatrix} = 0,$$

viz., this equation is satisfied if a denote any one of the quantities (σ, ρ, q, r) , and 1 any one of the same 4 quantities; that is, the equation of the surface is satisfied when (x, y, z, w) are the coordinates of a point on the line joining any two of the points.

a skew cubic; and on writing therein $P = 0, Q = 0$, the equations become

$$\begin{vmatrix} RS_1 + R_1S, & TU_1 + T_1U \\ RS, & TU \end{vmatrix} = 0,$$

viz., the line ($P = 0, Q = 0$) meets the skew cubic in the points where the line meets the quadric surfaces $RS_1 = 0, TU_1 = 0$ corresponding to it on the Jacobian.

90. To determine a skew cubic cutting twice each of 6 given lines, I proceed as follows. Let the lines be 1, 2, 3, 4, 5, 6 and correspondingly therein respectively the 10 skew cubics 1, 2, ..., 9, 10, where each of the skew cubics cuts the line 1, 2, taking the cubics except that denoted by the same number; we may, for instance, write down the 4 equations of the cubics, and from the equations of the 4 quadrics, the conditions among the 4 plane-pairs, since the points P and P' are on the same line through the vertex, P' are reciprocal to one another, P' to P , on which corresponds to the vertex, the corresponding ones to one another, we have therefore

$$J(P, T, U, V) = 0;$$

viz., the locus is the Jacobian of the foregoing definitions of the symmetroid. And the determinant gives in the equation expressed by the corresponds to the curve curves.

91. But the Jacobian may be obtained in a different manner; viz., if we establish the equations

$$(2P_1 + aQ_1 + bR_1)\delta_{1\nu} = 0,$$

$$(\quad)P_1 + \delta_{1\nu} = 0,$$

$$(\quad)Q_1 = 0,$$

$$(\quad)V = 0,$$

then the elimination of $(\xi, \eta, \zeta, \omega)$ from these is the equation of the Jacobian. And since each of the equations is symmetrical in regard to (x, y, z, w) , $(\xi, \eta, \zeta, \omega)$, the equation $J(P, T, U, V) = 0$ of the Jacobian is symmetrical in regard to (x, y, z, w) , $(\xi, \eta, \zeta, \omega)$.

the coefficients of U , V , we have between the points p , q , r a twofold relation, which may be represented as follows:

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ p & p & p & p & q & q & q \\ p & p & p & p & q & q & q \\ p & p & p & p & q & q & q \\ p & p & p & p & q & q & q \end{vmatrix} = 0,$$

it being understood that the p 's, q 's represent constants, and the p 's, q 's, r 's linear functions of these variables respectively. The two several equations of the system, regarding therein p , q , r as coordinates, represent each of them a quartic curve; any 2 of these are coresidual; and the number of points common to all the curves is $10 - 3 = 7$. There are corresponding included among the 10 points, but they are not in general included among the 10 points to be regarded as proper solutions; the others are extraneous solutions, accordingly the line $J(P, Q, R, V) = 0$ of the equations respectively; it is at once seen that any common solution to the foregoing equations corresponds to a value $p = q = r$ common to all the curves of the system; this solution counts, I believe, once only, and the number of relevant solutions is $7 - 1 = 6$.

92. It may be observed, in regard to the foregoing solution, that if we take $a = 0$, $b = 0$, $c = 0$ for the equation of the quadric surfaces through the points 1, 2, 3, and as the dianodal surfaces $U = 0$ and $V = 0$ may be taken to be other cones, the line through the singularity in the surfaces.

93. The conclusion is, that touching twice each of the six lines, we have one solution

$$\lambda 123 + \mu 124 + \nu 125 + \rho 126 = 0,$$

$$\lambda' 123 + \mu' 134 + \nu' 135 + \rho' 136 = 0,$$

respectively, the coefficients of the two surfaces being in evidence. And it is clear that for $\mu = \mu'$, $\nu = \nu'$, $\rho = \rho'$, the equations become respectively the equation of the quadric surface through the points 1, 2, 3, 4, and the equation of the quadric surface through the points 1, 2, 3, 4, that is the result.

94. The conclusion is, that touching twice each of the six lines 5, 6, 7, 8, 9, 10, we have 15 systems of lines 7, 8, 9, 10, and therefore 15 Jacobian surfaces.

Further Investigations as to the Jacobian, &c.

95. Taking $(\xi, \eta, \zeta, \omega)$ as plane-coordinates, the two quadric surfaces

$$(a, b, c, d, f, g, h, l, m \times \xi, \eta, \zeta, \omega)^2 = 0$$

and

$$(A, B, C, D, F, G, H, L, M, N \times \xi, \eta, \zeta, \omega)^2 = 0$$

are said to be intervex (or intervex) one of the other, when we have between the coefficients the relation

$$(a, b, c, d, f, g, h, l, m \times A, B, C, D, F, G, H, L, M, N) = 0,$$

that is

$$aA + \dots + 2fF + \dots = 0.$$

The condition that the two surfaces may be intervex of each other is linear in regard to the coefficients of each surface separately; hence, as before explained, intervex to two given quadrics σ quadrics; or generally, that intervex to $10 - k$ quadrics. And, moreover, if the quadrics be $L = 0, M = 0$, &c., of two given quadrics σ quadrics, we have between them the relation

$$(a, b, c, d, f, g, h, l, m, n \times L, M) = 0,$$

that is,

$$\alpha L + \dots = 0.$$

The condition that two surfaces are harmonic in regard to the pair of planes is linear in regard to the coefficients of each surface separately; hence, as before explained, intervex to two given quadrics σ quadrics; or generally, that intervex to $10 - k$ quadrics.

The Symmetroid in connection with the Jacobian.

96. If the equation $\nabla = 0$ of the symmetroid seems to contain homogeneously 40 constants. But starting from the only with the foregoing symmetrical determinant of the order 4 may multiply by a constant determinant, that is by a symmetrical determinant in the same way; or starting in such ways we may add to a symmetrical determinant. Their column or rows are interchanged by the determinant column, the constant determinants 40 constants; and the coefficients of the constant determinant may be used to be a constant determinant column; and the coefficients of the constant determinant may be used to be a symmetrical determinant column; or say each quartic surface $\nabla = 0$ of the form

Symmetroid with given Nodes.

97. A symmetroid can be formed with 7 given points as nodes; but there is no proper symmetroid with 8 given points as nodes. If we endeavour to form a symmetroid with 7 given points as nodes; if 8 nodes are required, then it is necessary that the resulting determinant must be the resulting condition that the 8th node may, however, be any point on the symmetroid; and the dianodal surface through the 7 points; and a 9th point anywhere on the dianodal curve through them; this will be probably nowhere on the surface. I have not succeeded in effecting this.

means of the surfaces 1 and 2 determine the two planes of the plane-pair, and then these will be harmonics in regard to the surface 3. It thus appears that the two planes 1, 2, 3, to the surfaces 1, 2, 4, the surfaces 1, 2, 5, &c. are intervex in regard to 1, 2, are the same as those which are intervex in regard to the involution in regard to the surfaces 1, 2, 6, 1, 2, 7, and we have therefore 6 nodes, since we always come round upon the surface (quoad the plane-pair).

98. It is substantially the same thing, but it is rather easier, to consider the whole question under the reciprocal form; viz., instead of a plane-pair and a quadric surface represented by the lines of intersection of the line $\nabla J(P, Q, R, V) = 0$ given quadric surfaces; viz., this is equivalent to the involution in regard to the given quadric surfaces, the involution in regard to the given quadric surfaces, viz., we have therein a line which is a (quoad the lines passes) in involution in regard to the three quadric surfaces. Conversely, when the axis of a plane-pair represented by the dianodal surface through the 9 points, considered as an involution in regard to the involution in regard to the 9 quadric surfaces.

99. Consider a line which is in involution 5 given quadric surfaces $(a, b, c, f, g, h, l, m, n \times P, Q, R, S, T)^2 = 0$; the plane P of the line through the same line through a 6th quadric surface, that we should have an indefinite series of identities of the form

$$(\alpha, \dots \times a, b, c, f, g, h, l, m, n \times P, Q, R, S, T) = 0;$$

viz., this expresses that the line a has, with the 5 given quadric surfaces, $aS + \beta T + \gamma U + \delta V + \epsilon W = 0$.

100. The numerical relations are as before. The transversal of any 6 lines, regarded as an involution 6 lines, is the dianodal surface through the 6 lines; and there are clearly 8 lines in involution to 8 quadric surfaces, the lines are intervex to 8 quadric surfaces, the lines are interpret as a system of $8 - 4 = 4$ lines; and similarly any 5 of the 8 quadric surfaces, the lines are 1, 2 of the 8 quadric surfaces. Let the equation of the 8th quadric surfaces be 7, 8, 9, 10, of the 5 quadric surfaces, the lines having $5 - 1 = 4$ involution in regard to 1, 1 of the 5 quadric surfaces have therefore the relation

$$\lambda 125 + \mu 124 + \nu 123 + \rho 126 = 0,$$

$$\lambda' 135 + \mu' 134 + \nu' 135 + \rho' 136 = 0,$$

respectively, the coefficients of the two surfaces being in evidence. And it is clear that for $\mu = \mu'$, $\nu = \nu'$, $\rho = \rho'$, the equation becomes respectively the equation of the quadric surface through the points 1, 2, 3, 4 and the equation of the quadric surface through the points 1, 2, 3, 4, that is the result.

101. The conclusion is, that touching twice each of the six lines, we have 15 systems of lines 7, 8, 9, 10, and therefore 15 Jacobian surfaces.

and conversely. But the value 20 (which represented without difficulty. For the question may be transformed as follows: It a single quartic surface having 5 given quadrics, how many of the axes of the lines of these (any) 4 are clear at any of those points one of the system there are quadrics, the same number which are connected to those quadrics, the same problem comes upon the form $\nabla = 0$, $\Theta = 0$ of the threefold relations

$$aS + \beta T + \gamma U + \delta V = 0, \quad (\sigma, \beta, \gamma, \delta) = 0,$$

where the coefficients $(\sigma, \beta, \gamma, \delta)$ are arbitrary; and the question becomes: how shall we have between the equations satisfied points common to the threefold relation $aS + \beta T + \gamma U + \delta V = 0$? we need not actually establish the equations; we have only to find a value λ such that we shall have between the equations satisfied points common to the three equations; we shall have

$$J(P, T, U, V) = 0;$$

and as the question becomes, what is the order of the system formed by the lines of intersection of the 4 quadric surfaces? But each pair of quadric surfaces will give a curve of the eighth order; and there are 6 such pairs, and the total order is $6 \cdot 8 = 48$. We have to subtract from this the order of the curves which lie on more than one such pair of quadric surfaces; we have to subtract from this the order of the curves which lie on more than one such pair of quadric surfaces; and the value obtained is 20.

102. Returning to the equation $\nabla = 0$ of the symmetroid, where ∇ is a symmetrical determinant of the order 4, we have between any two of the rows the relation

$$\nabla S + \nabla' T + \nabla'' U + \nabla''' V = \text{plane-pair};$$

viz., the values of $(\alpha, \beta, \gamma, \delta)$ which satisfy this condition belong to a threefold relation between the coefficients of aS , having previously explained location, so that the rows of the determinant have the threefold relations; and the the the threefold relation, the rows of the determinant have the property that, considering each line independent—the line $aS + \beta T + \gamma U + \delta V$ is, by the determinant relation, a plane-pair; and the threefold relations, the line is also a plane-pair.

103. In converse, in working this out, to consider H, K as standing, in the first instance, for $(x, y, z, w), (x', y', z', w')$, but in the other manner, w, w', z, z' , we have, with the constants a, b, c, d, e, f, g , &c., of the constant w, z , the constants a, c , of the constant. But what follows make no use of this last quartic.

104. Reverting to the equation $\nabla = 0$ of the symmetroid, where ∇ is a symmetrical determinant of the order 4 it has already explained, viz.,

$$(\sigma, \beta, \gamma, \delta) \times (\sigma', \beta', \gamma', \delta') \cdot xyz + \nu \sigma' \beta' \gamma' \delta' (yz - yz')^2 + \dots = 0,$$

where the second factor is

$$\begin{aligned} & (abc)(a'y' - y) + a'b'c \cdot c'a + a'b'c'(\sigma\delta' - \delta\sigma') + abc(\beta y' - y\beta') \\ & = b\sigma' b'\sigma(\sigma\delta + \beta\gamma) - z\sigma b'(\sigma\delta' + \beta\gamma') - z'\sigma' b(\sigma'\delta + \beta'\gamma) - z'\sigma'b'(\sigma'\delta' + \beta'\gamma'), \end{aligned}$$

or, omitting the factor $= ab$, the term in (xz) is $z(zb - h bz)$.

105. The final result contains 184 terms; viz., 6 terms in the multiplier a of aS, T, U, V, W , viz., $abc \cdot abf - hfbc + bcg \cdot abf$, &c., $abc \cdot abf$, &c. The final result may be exhibited in the same form

$$\begin{aligned} \Sigma = & + 2\{abc \cdot abf - hfbc + bcg \cdot abf - acf \cdot bgh + cgh \cdot bdh\} \\ & + 2\{bcd \cdot abf - lf cgh + df g \cdot bch - dgh \cdot abc + df g \cdot bch\} \\ & + 2\{efg \cdot abc + glg \cdot bcg + fch b - acg + chg \cdot bdh\} \\ & + 2\{bch \cdot abf + abc \cdot bgh - abc \cdot bgh + abc \cdot bgh\} \\ & + \dots \\ & = 0. \end{aligned}$$

Paragymmetrical Cyzo: the Hessian of a Cubic.

106. Reverting to the general equation

$$aS + \beta T + \gamma U + \delta V = \text{cone},$$

which connects the symmetroid and Jacobian, it is evident that if S, T, U, V are the derivatives, in regard to the coordinates, of a single cubic function, $U_c = (* \times x, y, z, w)^3$, then the symmetroid and the Jacobian become one and the same surface; viz., this is the Hessian surface derived from the given cubic surface. The two corresponding points P on the surface have a coincidence in this case; and the equation $\Phi = 0$ vanishes identically when S, T, U, V are the derivatives, in regard to the coordinates, of a single cubic surface; we obtain

$$\nabla = \begin{vmatrix} S_x & T_x & U_x & V_x \\ S_y & T_y & U_y & V_y \\ S_z & T_z & U_z & V_z \\ S_w & T_w & U_w & V_w \end{vmatrix} = 0,$$

which is the Hessian; and the equation may equally be regarded as a Jacobian equation

$$J(S, T, U, V) = 0,$$

a form which puts in evidence the properties just referred to.

Quartics with 11 or more Nodes.

107. I mention two results which, although they relate to quartic surfaces with more than 10 nodes, present themselves in such immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A, & H, & G, & L \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \dots$ linear functions of the coordinates), we have identically $A = 0$, then the surface has a node $H = 0, G = 0, L = 0$; or the surface has in all 11 nodes. And so also if (identically in every case) B is $= 0$, there are 12 nodes; if C is $= 0$, there are 13 nodes; and if D is $= 0$, 14 nodes respectively, mentioned in Kummer's Memoir.

108. We may consider the symmetroid derived from the quartic surfaces which pass through 6 given points; viz., taking as before (see No. 25) the coordinates of the 6 points, present themselves in the same immediate connexion with the present Memoir, that it is natural to speak of them. If, in the equation

$$\begin{vmatrix} A, & H, & G, & M \\ H, & B, & F, & M \\ G, & F, & C, & N \\ L, & M, & N, & D \end{vmatrix} = 0$$

of the symmetroid ($A, B, \dots$ linear functions of the coordinates), we have identically $H = 0$ (and not identically) every case) A is $= 0$, there are 11 nodes; if B is $= 0$ (and not identically) every case) A is $= 0$ there are 12 nodes; if C is $= 0$, there are 13 nodes; and if D is $= 0$, there are 14 nodes respectively, mentioned in Kummer's Memoir with 11, 12, 13, and 14 nodes respectively. But I reserve this for future consideration.

Quartic Surface through three given Lines.

109. I take the opportunity of mentioning some results which have a connexion, although not an immediate one, with the subject of the present Memoir.

Quartic Surface through three given Lines.

110. To find the equation to the quartic surface through the three lines

$$(a_1, b_1, c_1, f_1, g_1, h_1), \quad (a_2, b_2, c_2, f_2, g_2, h_2), \quad (\alpha', \beta', \gamma', \delta'),$$

the points $(\sigma, \beta, \gamma, \delta)$, $(\sigma', \beta', \gamma', \delta')$ the coordinates of a quadric surface through this line are of the form

$$\begin{vmatrix} x^4 & y^4 & z^4 & w^4 & yz & xz & xy & xw & yw & zw \\ x^3y & xy^3 & yz^3 & y^3z & yzw & xzw & xyw & xyz & y^2zw & yz^2w \\ 2x^3 & 2xy & 2yz & 2zw & a\beta' & a\gamma' & a\delta' + a\beta'\gamma' & b\gamma' + a\beta'\delta' & c\delta' + a\gamma'\beta' & d \\ \sigma^4 & \beta^4 & \gamma^4 & \delta^4 & \beta\gamma & a\sigma & a\delta & a\gamma & \beta\delta & \gamma\delta \\ \sigma'^4 & \beta'^4 & \gamma'^4 & \delta'^4 & \beta'\gamma' & a'\beta' & a'\delta' & a'\gamma' & \beta'\delta' & \gamma'\delta' \end{vmatrix} = 0;$$

and if we form thus a determinant with three of its lines relating to the line 1, three to the line 2, and three to the line 3, we have the equation of the quartic surface through the three lines 1, 2, 3.

Condition that five given lines may lie on a Cubic Surface.

111. Taking the lines to be $(a, b, c, f, g, h)_1, \dots, (a, b, c, f, g, h)_5$, and (x, y, z, w) the coordinates of any point on the line, the equation of the cubic surface through the 5 lines, taken to be $(1, b_1, c_1, c'_1), \dots, (1, b_5, c_5, c'_5)$; and the equation of the Jacobian surface of the 5 points.

112. The complete development of the equations above written, in the form best suited for actual application, is a calculation of some length, and would require to be performed with considerable care. The other terms were obtained therefrom by changing a, b into $a + \frac{1}{w}(\beta y - p)$, $b + \frac{1}{w}(\alpha \delta - p\beta)$, respectively; the equation may therefore be written in the symbolic form

$$\omega^4 \cdot \exp \frac{1}{\omega} [h(bg - p)\delta_a + (-h\sigma + fz)\delta_b + (p\omega - fp)\delta_c] \cdot [(a, b, c, d)^4] = 0,$$

or, what is the same thing,

$$\omega^4 \cdot \exp \frac{1}{\omega} [a\{ph(p\beta - bh\delta) + g(hb\delta - p\beta_1) + z(p\beta_2 - pb)\}\delta_a + \dots] = 0,$$

where $\exp$ (read exponential) denotes e^z , and $\{(a, b, c, d)\}$ represents a determinant as above explained. The equation contains, it is clear, the four terms

$$\omega^4 [(a, -b, p)\beta_2 + y(b, -h, p)\delta_a\beta + \dots + z'(c, \delta_a c'') + \omega'(a, b, c'')] = 0.$$

I am not sure whether the surface of the eighth order has been anywhere considered.

[End of paper 445.]